

Reciprocal Covariance Pencils and Equal-Specific-Damping Symmetry in Two-Mode Inertial Ornstein–Uhlenbeck Systems

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Abstract

We study a two-mode inertial (position–momentum) Ornstein–Uhlenbeck system driven by two independent thermal baths, one coupled to each oscillator’s momentum coordinate. Writing C_1, C_2 for the stationary covariance matrices produced by each bath acting alone, the joint stationary covariance is the affine pencil $\Sigma(T_1, T_2) = T_1 C_1 + T_2 C_2$, and for the bath-scale ratio $s = T_1/T_2$ the quartic $P(s) = \det(sC_1 + C_2)$ governs whether exchanging the two bath scales ($s \mapsto 1/s$) induces a reciprocal symmetry of the stationary covariance determinant. We show that, on the genuinely coupled interior of parameter space, $P(s)$ is palindromic if and only if the two oscillators share equal specific damping $\delta_i = \gamma_i/m_i$, equivalently $\gamma_1/m_1 = \gamma_2/m_2$. This is obtained from a stronger, oriented result: the log-determinant comparator $F = \log(\det C_2 / \det C_1)$ is strictly monotone in δ_1 at fixed $\omega_1, \omega_2, k, \delta_2$, so that $\text{sign } F = -\text{sign}(\delta_1 - \delta_2)$ throughout the admissible domain, with a single zero crossing exactly at $\delta_1 = \delta_2$. At that balance locus we further establish the exact identity $\partial F / \partial \delta_1|_{\delta_1=\delta_2=\delta} = -4/\delta$, independent of the oscillator frequencies and coupling strength. We give the exact positivity certificate underlying global monotonicity, record an equivalent reciprocal-generalized-eigenvalue characterization of palindromicity via $C_1^{-1}C_2$, and reduce the palindromic quartic to an explicit quadratic in $w = s + s^{-1}$ on the balance locus. Several of the polynomial identities used below were established by exact computer algebra (symbolic simplification to zero and zero-remainder polynomial division, never floating-point sampling); this is disclosed explicitly where it occurs, together with the accompanying verification scripts.

1 Introduction

A first-order (overdamped) pair of coupled Ornstein–Uhlenbeck processes driven by two independent baths produces a stationary covariance whose determinant, viewed as a function of the bath-scale ratio s , is quadratic in s ; on the relevant reciprocal locus it reduces to an affine relation in $w = s + s^{-1}$. That first-order theory is developed separately [2] and formalized in this repository’s companion Lean development.

Here we ask the analogous question for the *inertial* extension of that system: two damped harmonic oscillators, each carrying its own position and momentum coordinate, linearly coupled through a shared spring constant, each driven by an independent thermal bath acting on its own momentum coordinate. The stationary covariance is now a 4×4 matrix, and the bath-pencil determinant $P(s) = \det(sC_1 + C_2)$ is a genuine quartic in s . We ask exactly when this quartic is palindromic – equivalently, when its four roots (necessarily real and positive, in a sense made precise below) close under $s \mapsto 1/s$.

The answer is that the criterion is *equal specific damping* $\delta_1 = \delta_2$, $\delta_i := \gamma_i/m_i$ – not equal raw damping constants γ_i and not equal masses m_i individually. We prove this as an unconditional *if and only if* statement on the genuinely-coupled interior of parameter space (Theorem 2), via a stronger oriented result (Theorem 1) giving the sign of the log-determinant comparator $F = \log(\det C_2/\det C_1)$ everywhere, not merely at the palindromic locus. A targeted literature search did not locate an exact prior formulation of this criterion, the oriented comparator, or the balance-slope identity of Proposition 1; the underlying machinery – stationary covariances of linear systems via the Lyapunov equation [3], palindromic-polynomial algebra, and structured matrix-pencil eigenvalue theory [4] – is classical or a known tool, cited where it is used below. We report the absence of an exact match as a limitation of the search performed, not as a claim of priority.

We emphasize at the outset that the change in polynomial degree between the first-order case (quadratic in s , affine in w) and the inertial case (quartic in s , quadratic in w on the reciprocal locus) is a computed fact about each specific model, not an instance of a general “dynamical-order law” relating state dimension to reciprocal-polynomial degree; no such general claim is made or needed here.

2 Model

Consider two damped harmonic oscillators with masses m_1, m_2 , natural frequencies ω_1, ω_2 (in the original, mass-bearing coordinates), raw damping constants γ_1, γ_2 , and a linear spring coupling of strength k . Each oscillator’s momentum coordinate is driven by an independent white-noise bath. The full state is $(x_1, p_1, x_2, p_2) \in \mathbb{R}^4$.

Mass-weighted reduction. The canonical transformation $(x_i, p_i) \mapsto (\sqrt{m_i} x_i, p_i/\sqrt{m_i})$ has Jacobian determinant exactly 1, and reduces the general-mass problem to an equivalent *unit-mass* problem depending on the masses and raw parameters only through the *specific damping* rates

$$\delta_i := \gamma_i/m_i,$$

together with reduced frequencies absorbing the $1/m_i$ factors. Because the transformation has unit Jacobian, it preserves every covariance determinant appearing below; the palindromicity criterion we prove is therefore exactly a statement about δ_1, δ_2 , not about γ_i or m_i individually. From here on, ω_1, ω_2, k denote the (already reduced) unit-mass parameters, and we work entirely in the unit-mass system; the final theorem is restated in terms of γ_i/m_i in Section 9.1.

Drift and diffusion. In the unit-mass system, with state ordering (x_1, p_1, x_2, p_2) ,

$$A = \begin{pmatrix} 0 & 1 & 0 & 0 \\ -(\omega_1^2 + k) & -\delta_1 & k & 0 \\ 0 & 0 & 0 & 1 \\ k & 0 & -(\omega_2^2 + k) & -\delta_2 \end{pmatrix}, \quad Q_1 = \text{diag}(0, 2\delta_1, 0, 0), \quad Q_2 = \text{diag}(0, 0, 0, 2\delta_2).$$

Q_1, Q_2 are the diffusion matrices for bath 1 and bath 2 acting alone (unit temperature; a bath at temperature T_i scales Q_i by T_i , and the full joint stationary covariance under both baths simultaneously, $\Sigma(T_1, T_2)$, is linear in T_1, T_2 – see Section 3). The stationary covariances C_1, C_2 of the two single-bath problems solve the Lyapunov equations

$$AC_i + C_i A^\top = -Q_i, \quad i = 1, 2. \tag{1}$$

Relation to prior models. Two damped harmonic oscillators, each carrying its own position and momentum coordinate and each driven by an independent heat bath, is not a new model class: the same 4×4 Lyapunov-equation setup (uniform damping, in that case) was used by Boffi and De Gregorio [1] to study a resonance phenomenon in the covariance entries as the ratio of the two spring constants varies. Their analysis fixes a common damping rate throughout and does not address palindromicity, reciprocal spectra, or any damping-ratio criterion; the present work instead varies the two specific damping rates δ_1, δ_2 independently and asks exactly when the resulting covariance-pencil determinant is palindromic.

Admissible domain. Throughout, $\omega_1, \omega_2, k, \delta_1, \delta_2 > 0$ (open, all strict). The sole boundary degeneracy is $k = 0$: at $k = 0$ the drift matrix A becomes exactly block-diagonal, bath 1 cannot influence oscillator 2 (and vice versa), and C_1, C_2 become rank-deficient (singular). For $k \neq 0$, the single-bath Kalman/Krylov controllability determinant for each channel equals k^2 exactly, so $C_1, C_2 \succ 0$ for every $k > 0$ (Lemma 1). No further restriction on any parameter is used anywhere below.

3 Bath-response covariances and the pencil

Lemma 1 (Pencil and positivity). *With $Q(T_1, T_2) := T_1 Q_1 + T_2 Q_2$ and $\Sigma(T_1, T_2)$ the stationary solution of $A\Sigma + \Sigma A^\top = -Q(T_1, T_2)$,*

$$\Sigma(T_1, T_2) = T_1 C_1 + T_2 C_2.$$

Writing $s := T_1/T_2$ and $P(s) := \det(sC_1 + C_2)$,

$$P(s) = c_4 s^4 + c_3 s^3 + c_2 s^2 + c_1 s + c_0, \quad c_4 = \det C_1, \quad c_0 = \det C_2.$$

For $k > 0$, $C_1 \succ 0$ and $C_2 \succ 0$.

Proof sketch. Linearity of Σ in (T_1, T_2) follows from linearity of the Lyapunov equation (1) in the forcing term. c_4, c_0 are the s^4, s^0 coefficients of $\det(sC_1 + C_2)$, which equal $\det C_1, \det C_2$ by multilinearity of the determinant in the pencil parameter. Positive definiteness for $k > 0$ follows from the exact controllability determinant k^2 (single-bath Kalman rank criterion applied to (1)) together with A 's stability for every $\delta_1, \delta_2 > 0$: a stable, controllable pair yields a positive definite stationary Gramian, a classical fact of linear systems theory [3]. $\square$

4 Palindromic sufficiency

Lemma 2 (Sufficiency). *If $\delta_1 = \delta_2$ then $P(s)$ is palindromic: $c_4 = c_0$ and $c_3 = c_1$, for every admissible $\omega_1, \omega_2, k > 0$.*

The shortest available proof substitutes $\delta_1 = \delta_2 = \delta$ directly into (1) before solving. This is a genuine structural simplification, not merely a reduction in the number of free symbols: with common specific damping, the p -block of A 's damping is proportional to the identity, hence commutes with any orthogonal transform diagonalizing the stiffness block $\begin{pmatrix} \omega_1^2 + k & -k \\ -k & \omega_2^2 + k \end{pmatrix}$. This decouples the deterministic drift into two independent normal modes with frequencies Ω_1, Ω_2 (the stiffness block's eigenvalues) and common damping δ , while the per-bath noise becomes a rank-one source correlated across the two modes – the same structure used for Proposition 1 below. Solving (1) in these coordinates gives $C_1(\delta), C_2(\delta)$ in closed form, and $c_4 - c_0 = 0, c_3 - c_1 = 0$ are then confirmed by exact symbolic simplification (Section 13, item 1).

5 The log-determinant comparator

Definition 1. $F(\delta_1, \delta_2; \omega_1, \omega_2, k) := \log(\det C_2 / \det C_1)$, well-defined on the admissible interior by Lemma 1.

$F = 0$ is exactly bath-response volume balance ($\det C_1 = \det C_2$), and sign F distinguishes the two possible orderings of the two single-bath covariance volumes. By Lemma 2, $F(\delta_2, \delta_2) = 0$ for every admissible $\omega_1, \omega_2, k, \delta_2$ – the base point used throughout Sections 6–8.

6 Universal balance slope

Proposition 1 (Balance-locus slope). *At $\delta_1 = \delta_2 = \delta$,*

$$\left. \frac{\partial F}{\partial \delta_1} \right|_{\delta_1 = \delta_2 = \delta} = -\frac{4}{\delta},$$

for every admissible $\omega_1, \omega_2, k, \delta > 0$ – independent of the oscillator frequencies and the coupling strength.

Proof sketch. Using the normal-mode decoupling of Section 4, the adjoint-Lyapunov sensitivity identity gives, for $X_i := \partial C_i / \partial \delta_1$,

$$\frac{\partial F}{\partial \delta_1} = \text{tr}(C_2^{-1} X_2) - \text{tr}(C_1^{-1} X_1) = 2 \left[e_2^\top Y_2 v_2 - e_2^\top Y_1 u_1 \right],$$

where $e_2 = (0, 1, 0, 0)^\top$, $u_1 = C_1 e_2 - e_2$, $v_2 = C_2 e_2$, and Y_i solves $A^\top Y_i + Y_i A = C_i^{-1}$ (this sign – positive, not negative – was fixed after an initial sign error was caught by cross-checking against direct differentiation; see Section 13). Substituting the closed-form $C_1(\delta), C_2(\delta)$ from Section 4 and solving for Y_1, Y_2 symbolically (fully general in $\omega_1, \omega_2, k, \delta$) yields the stated closed form after exact simplification. The ω_1, ω_2, k -cancellation arises from a combination of (i) the equipartition structure of each normal mode’s self-covariance, (ii) the two channels sharing the identical denominator $k^4 / (\Delta D_{12}^2)$ at $\delta_1 = \delta_2$ (with Δ, D_{12} as in Section 7), which cancels in the derivative ratio, and (iii) the shared specific damping making the rank-one sensitivity $A' = -e_2 e_2^\top$ act identically on both channels’ drift operator. This is a proved structural fact about this specific model class, not claimed to generalize beyond it. $\square$

7 Global monotonicity

Lemma 3 (Derivative factorization).

$$\frac{\partial F}{\partial \delta_1} = \frac{K_{\text{total}}(\omega_1, \omega_2, k) \cdot \delta_1^3 \delta_2^4}{K(\delta_1, \delta_2)^4 \cdot \det C_1 \cdot \det C_2},$$

where

$$K(\delta_1, \delta_2) = a_1 \delta_1^3 \delta_2 + a_2 \delta_1^2 \delta_2^2 + a_3 \delta_1^2 + a_4 \delta_1 \delta_2^3 + a_5 \delta_1 \delta_2 + a_6 \delta_2^2,$$

$$a_1 = \omega_2^2 + k, \quad a_2 = \omega_1^2 + \omega_2^2 + 2k, \quad a_3 = k^2, \quad a_4 = \omega_1^2 + k, \quad a_5 = (\omega_1^2 - \omega_2^2)^2 + 2k^2, \quad a_6 = k^2,$$

and

$$K_{\text{total}}(\omega_1, \omega_2, k) = -\frac{4k^8}{\Delta^2}, \quad \Delta := k\omega_1^2 + k\omega_2^2 + \omega_1^2 \omega_2^2.$$

Proof sketch. The six-monomial support and coefficient functions $a_1, \dots, a_6$ were obtained by exact-rational interpolation across many (ω_1, ω_2, k) configurations and then proved as an identity by exact zero-remainder polynomial division at independent, freshly chosen points (Section 13, item 2). K_{total} is determined by evaluating the identity at the (fully general, symbolic) balance locus $\delta_1 = \delta_2 = \delta$, where $K(\delta, \delta) = \delta^2 D_{12}$ with $D_{12} = (\Omega_1^2 - \Omega_2^2)^2 + 2\delta^2(\Omega_1^2 + \Omega_2^2)$ the normal-mode cross-Sylvester-resolvent denominator of Section 4, $\Delta = \Omega_1^2 \Omega_2^2$ the product of squared normal-mode frequencies, and $\det C_1 = \det C_2 = k^4/(\Delta D_{12}^2)$ there; combined with Proposition 1's closed form $F' = -4/\delta$, this determines K_{total} uniquely (it is independent of δ by construction, so any one evaluation point suffices), and the result is independently re-confirmed by exact zero-remainder division against the general (non-symmetric) identity at several fresh points, including off-diagonal ones (Section 13, item 3). $\square$

Theorem 1 (Global monotonicity and oriented comparator). *Each $a_1, \dots, a_6 > 0$ by inspection (sums of squares and manifestly positive terms; $a_5 > 0$ even at $\omega_1 = \omega_2$, since $k > 0$), so $K > 0$ for all $\delta_1, \delta_2 > 0$ (a sum of six positive terms). $\Delta > 0$ (a sum of three positive terms), so $K_{\text{total}} < 0$. Since $\det C_1, \det C_2 > 0$ (Lemma 1), every factor in Lemma 3's identity has manifest sign, giving*

$$\frac{\partial F}{\partial \delta_1} < 0$$

for every admissible interior point (fixed $\omega_1, \omega_2, k, \delta_2$). Combined with the base point $F(\delta_2, \delta_2) = 0$ (Section 4), F is strictly decreasing in δ_1 with a single zero crossing at $\delta_1 = \delta_2$:

$$\text{sign } F = -\text{sign}(\delta_1 - \delta_2).$$

Monotonicity of Gramian-type quantities under parameter perturbations is a recurring theme in linear systems and sensitivity theory; we did not locate a general theorem in the literature searched that directly implies this specific determinant-ratio monotonicity for the pencil $sC_1 + C_2$, so we record it here as established for this model by the explicit factorization of Lemma 3 rather than as an instance of a known general result.

8 Oriented volume theorem

Corollary 1 (Unique volume balance).

$$\det C_1 = \det C_2 \iff \delta_1 = \delta_2.$$

This is immediate from Theorem 1 and Definition 1 ($(F = 0 \iff \det C_1 = \det C_2)$), and is a central result in its own right, not merely a device toward palindromicity: it establishes that the bath-response covariance *volumes* $\det C_1, \det C_2$ coincide exactly once, at exactly the equal-specific-damping locus, with a definite orientation on either side.

9 Global palindromicity theorem

Theorem 2 (Palindromicity iff equal specific damping). *For the admissible genuinely coupled two-mode inertial OU system,*

$$P(s) \text{ is palindromic} \iff \delta_1 = \delta_2.$$

Proof. Necessity. If $P(s)$ is palindromic then $c_4 = c_0$, i.e. $\det C_1 = \det C_2$ (Lemma 1), hence $\delta_1 = \delta_2$ (Corollary 1). The coefficient identity $c_3 = c_1$ is not invoked. *Sufficiency.* Lemma 2. $\square$

9.1 Restoring general masses

Restoring $\delta_i = \gamma_i/m_i$ (Section 2; the mass-weighted reduction is exact and preserves every determinant used above):

$$P(s) \text{ is palindromic} \iff \frac{\gamma_1}{m_1} = \frac{\gamma_2}{m_2}.$$

10 Spectral interpretation

Corollary 2 (Reciprocal generalized eigenvalues). *For $k > 0$, $N := C_1^{-1}C_2$ is similar to a symmetric positive-definite matrix (via $C_1^{-1/2}$), hence has four strictly positive real eigenvalues $\mu_1, \dots, \mu_4$ – the roots of $P(s) = 0$ are $s = -\mu_i$. $P(s)$ is palindromic if and only if $\{\mu_1, \dots, \mu_4\}$ splits into two reciprocal pairs $\{\mu, 1/\mu\}, \{\nu, 1/\nu\}$. Combined with Theorem 2:*

$$\delta_1 = \delta_2 \iff P(s) \text{ palindromic} \iff \{\mu_i\} \text{ splits into reciprocal pairs.}$$

We emphasize the convention: eigenvalues are those of $C_1^{-1}C_2$ (equivalently, of the symmetrized matrix), not $C_2^{-1}C_1$; the sign convention $s = -\mu_i$ in the palindromic-root characterization follows from writing $P(s) = \det C_1 \cdot \det(sI + C_1^{-1}C_2)$. Reciprocal eigenvalue pairing for structurally palindromic matrix pencils (those built from an explicit relation such as $B = A^\top$) is classical matrix-pencil theory [4]; the pencil $sC_1 + C_2$ studied here carries no such a priori structural relation between C_1 and C_2 (each solves an independent Lyapunov equation), so the reciprocal pairing recorded in this corollary is a *derived* consequence of Theorem 2, not an instance of the structural theory.

11 Reciprocal coordinate

On the palindromic locus ($\delta_1 = \delta_2$), dividing $P(s)$ by s^2 and using the classical reciprocal-equation identity $s^2 + s^{-2} = w^2 - 2$ with $w := s + s^{-1}$,

$$P(s)/s^2 = c_4 w^2 + c_3 w + (c_2 - 2c_4),$$

a genuine (non-degenerate) quadratic in w , with c_4, c_3, c_2 explicit rational functions of $\omega_1, \omega_2, k, \delta$. This is the inertial analogue of the affine reciprocal-coordinate reduction in the first-order OU problem [2]; the change in degree (affine versus quadratic) is a fact about the two specific models, not an instance of a general law relating state dimension to reduced-polynomial degree.

12 Discussion

Specific damping, not raw parameters, controls reciprocal symmetry. The palindromicity criterion is $\gamma_1/m_1 = \gamma_2/m_2$, not $\gamma_1 = \gamma_2$ or $m_1 = m_2$ individually – a direct consequence of the mass-weighted reduction (Section 2) having unit Jacobian. *Volume balance is unique* (Corollary 1): the two single-bath covariance determinants coincide at exactly one locus. *The comparator is oriented:* F does more than vanish at balance – its sign tracks which bath produces the larger stationary covariance volume, throughout the whole domain (Theorem 1). *The balance crossing has a universal slope*, $-4/\delta$, independent of frequencies and coupling (Proposition 1). *Reciprocal spectral pairing* (Section 7) gives a coordinate-independent restatement of the same criterion. The decomposition of the monotonicity identity into canonical internal response terms ($F' = 2(R_2 - R_1)$ in the notation of Proposition 1's proof) may provide a direction for future work.

12.1 Relation to the first-order theory

The first-order OU reciprocity result [2] (see also this repository’s Lean formalization) concerns a 2×2 stationary covariance, a quadratic pencil determinant, a trace-normalized forcing-ratio criterion, and an affine reduction in w . The inertial system studied here has a 4×4 covariance, a quartic pencil determinant, an equal-specific-damping criterion, a quadratic w -reduction on the reciprocal locus, and the additional oriented-volume theorem with no first-order analogue established here. The inertial result is not a routine extension of the first-order one: the palindromicity criterion, the reduced polynomial’s degree, and the proof architecture (normal-mode decoupling, adjoint-Lyapunov sensitivity) are each specific to the inertial state-space structure.

12.2 Limitations

This work concerns a linear Gaussian system with exactly two inertial modes, stationary statistics only, and the specific per-bath forcing structure of Section 2 (each bath coupled to exactly one oscillator’s momentum coordinate). Results are established on the genuinely coupled interior $k > 0$; no claim is made at $k = 0$. No claim is made about nonlinear extensions, about networks of more than two oscillators, or that the exact slope $-4/\delta$ holds outside this specific model class.

13 Computer-assisted proof disclosure

Several polynomial identities used above were verified by exact computer algebra. These checks use symbolic or exact-rational arithmetic rather than floating-point sampling; the accompanying verification scripts (`verification/inertial/` in the project repository) reproduce every load-bearing identity from a single documented entry point, `verify_global_chain.py`.

1. **Sufficiency** (Lemma 2): exact symbolic computation – $C_1(\delta), C_2(\delta)$ solved via exact linear algebra with $\delta_1 = \delta_2 = \delta$ substituted before solving; $c_4 - c_0, c_3 - c_1$ confirmed to simplify symbolically to 0. (`verify_sufficiency.py`)
2. **The six-coefficient K identity** (Lemma 3): discovered via exact-rational interpolation, proved by exact zero-remainder polynomial division against the independently computed true numerator, at fresh points not used in the interpolation.
3. $K_{\text{total}} = -4k^8/\Delta^2$ (Lemma 3): derived from the balance-slope proposition, independently re-confirmed by exact zero-remainder division at fresh points including an off-diagonal ($\delta_1 \neq \delta_2$) end-to-end check. (`verify_general_Fprime.py`)
4. **The balance-slope proposition** (Proposition 1): fully symbolic exact derivation (not sampled) via the adjoint-Lyapunov route, with the sign convention on the Y_i equation independently cross-checked against direct symbolic differentiation. (`verify_balance_slope.py`)

Given these closed forms, $K > 0$, $K_{\text{total}} < 0$, $F' < 0$, the oriented sign theorem, and palindromicity necessity are each short, human-checkable steps (Section 7–9). Earlier stages of this investigation used large parameter sweeps (a 729-point rational search, 215 and 102 exact line searches, 17 six-monomial discovery configurations, 66 scalar-sign spot checks) to locate the closed forms above before they were proved in general; that search history carries no theorem weight and is not part of this proof.

14 Reproducibility

All symbolic computation uses SymPy with exact (arbitrary-precision integer and rational) arithmetic; no floating-point computation is used in any load-bearing step. The verification workflow is:

```
python verification/inertial/verify_global_chain.py
```

which reproduces, from a single command, every identity disclosed in Section 13 together with a fresh end-to-end regression check (new parameter points, not used in discovery) confirming positive-definiteness, the base point, global monotonicity, the oriented sign theorem, and palindromicity exactly at $\delta_1 = \delta_2$ together with its exact failure at representative $\delta_1 \neq \delta_2$ points. Proof-generating and verification scripts (`verification/inertial/`) are kept separate from this project’s historical exploratory search scripts, which are not included in this repository.

AI assistance disclosure

The research and conceptual development are by Elias De Jesús, with AI assistance used in symbolic-workflow development, code generation and debugging, adversarial proof auditing, and manuscript editing. Mathematical claims in this manuscript were checked through the documented exact computational workflow of Section 13 and Section 9; independent human derivation is not claimed for the identities disclosed there as computer-assisted.

Acknowledgments

None.

References

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